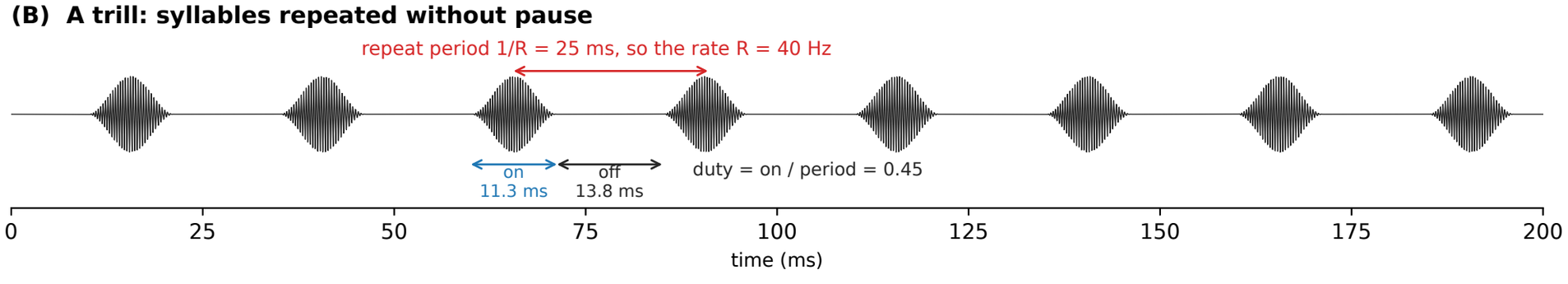
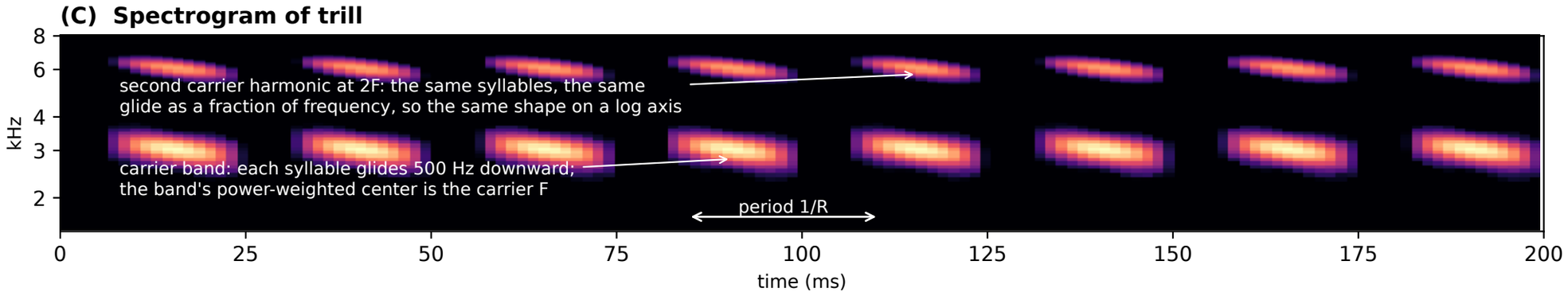
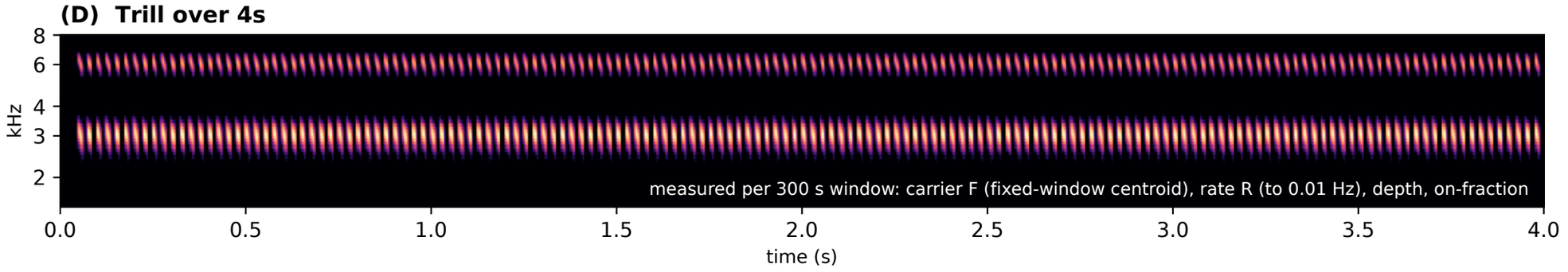
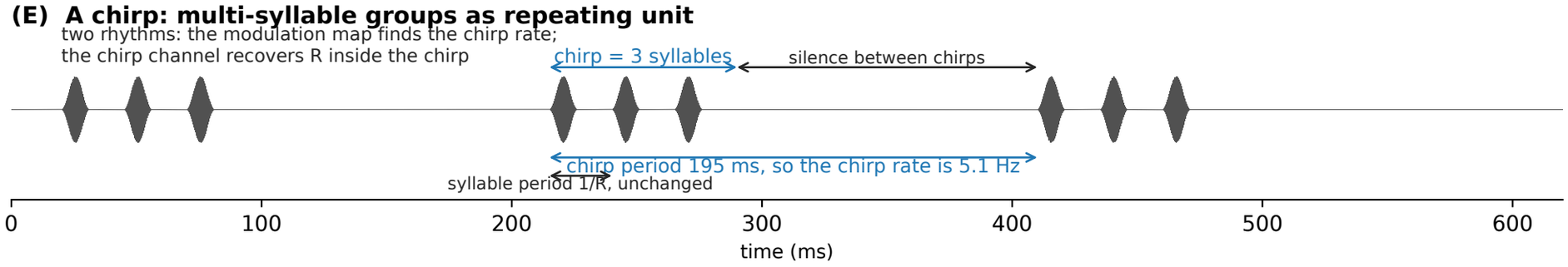
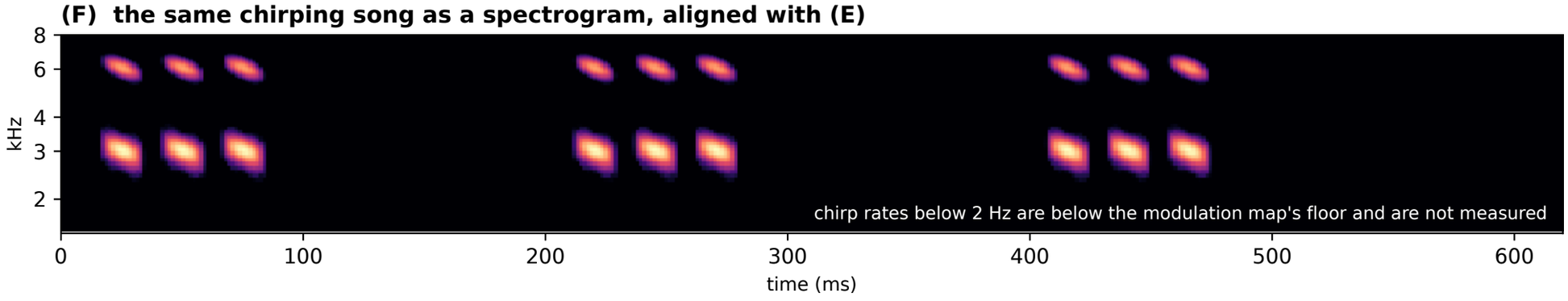

(A) A syllable
one cycle of the carrier: 1/F = 0.33 ms (F = 3000 Hz)
envelope
(amplitude modulation)
syllable (pulse) duration, 11.3 ms: one closing stroke of the wings
0
2
4
6
8
10
12
14
16
time (ms)

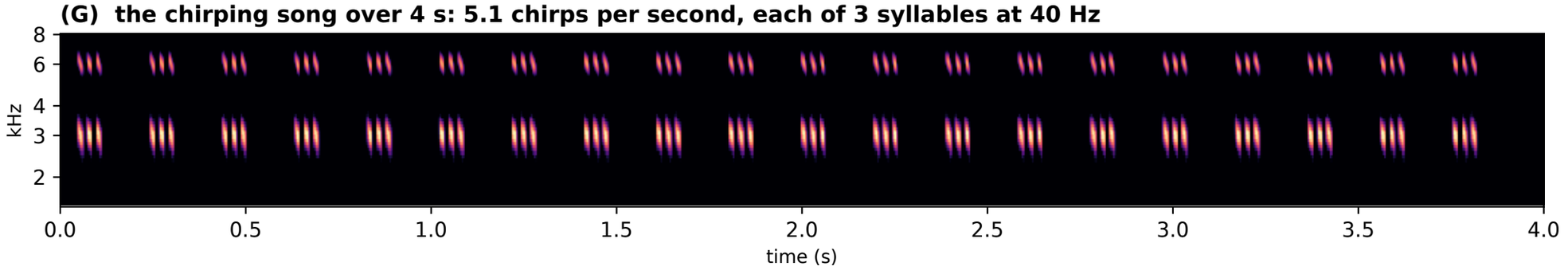

(B) A trill: syllables repeated without pause
repeat period 1/R = 25 ms, so the rate R = 40 Hz
on
11.3 ms
off
13.8 ms
duty = on / period = 0.45
0
25
50
75
100
125
150
175
200
time (ms)


(C) Spectrogram of trill
kHz
second carrier harmonic at 2F: the same syllables, the same
glide as a fraction of frequency, so the same shape on a log axis
carrier band: each syllable glides 500 Hz downward;
the band's power-weighted center is the carrier F
period 1/R
time (ms)


(D) Trill over 4s
kHz
measured per 300 s window: carrier F (fixed-window centroid), rate R (to 0.01 Hz), depth, on-fraction
time (s)


(E) A chirp: multi-syllable groups as repeating unit
two rhythms: the modulation map finds the chirp rate;
the chirp channel recovers R inside the chirp
chirp = 3 syllables
silence between chirps
chirp period 195 ms, so the chirp rate is 5.1 Hz
syllable period 1/R, unchanged
time (ms)


(F) the same chirping song as a spectrogram, aligned with (E)
kHz
chirp rates below 2 Hz are below the modulation map's floor and are not measured
time (ms)


(G) the chirping song over 4 s: 5.1 chirps per second, each of 3 syllables at 40 Hz
kHz
time (s)